TOPSTECHNOLOGIES
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Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Network devices

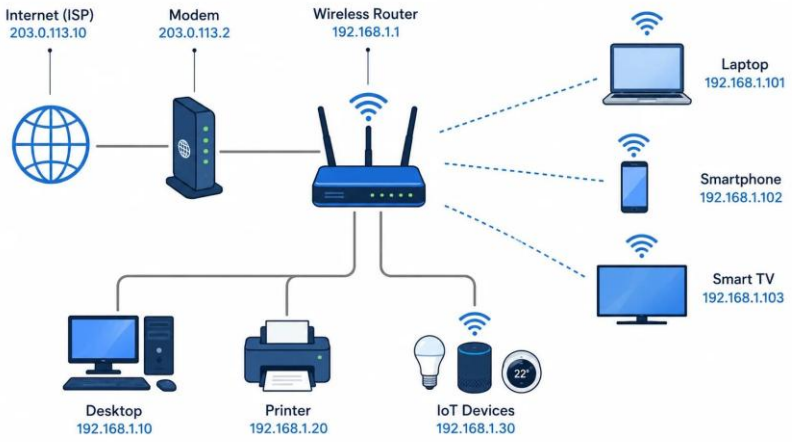
Task:

❖ **1. Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.**

➢ **Introduction**

- A home network is a group of devices connected to share an internet connection and communicate with each other. A router, switch, and network-enabled devices work together to provide reliable connectivity at home.

➢ **Home Network Diagram**



- **Explanation of Network Devices**

  1. Router: A router connects the home network to the Internet and directs data between networks. A Wi-Fi router also allows wireless devices to connect to the network.

  2. Switch: A network switch connects multiple wired devices within the same local area network (LAN). It forwards data to the appropriate connected device.

  3. Laptop: A laptop can connect to the network through an Ethernet cable or Wi-Fi. It is used for browsing, online classes, and file sharing.

  4. Smartphone: A smartphone usually connects to the home network through Wi-Fi. It can access the Internet, stream videos, and use online applications.

  5. Smart TV: A smart TV can connect through Wi-Fi or an Ethernet cable. It is used for streaming movies, online videos, and live sports.

- **Working of the Home Network**

  1. The Internet service enters the home through the Internet connection provided by the Internet Service Provider (ISP).
  2. The router connects the home network to the Internet.
  3. The switch connects multiple wired devices to the local network.
  4. The laptop can connect to the switch through an Ethernet cable or directly to the router.
  5. The smartphone connects wirelessly to the Wi-Fi router.
  6. The smart TV connects through Wi-Fi or Ethernet to access online streaming services.

❖ **2. Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).**

➢ **Introduction**

- Network devices are hardware components used to connect computers, smartphones, and other devices so that they can communicate and share network resources. Different network devices perform different functions, such as connecting networks, forwarding data, and providing wireless connectivity.

➢ **Comparison Table of Network Devices**

Network Device	Main Function	Real-World Use Case
Router	Connects different networks and directs data packets to their destinations.	A home Wi-Fi router provides Internet access for watching YouTube videos, using WhatsApp, and attending online classes.
Switch	Connects multiple wired devices within a Local Area Network (LAN) and forwards data to the intended device.	In a computer lab, a switch connects multiple computers for file sharing, accessing college resources, and playing multiplayer LAN games.

Hub	Connects multiple devices in a LAN and broadcasts incoming data to all connected ports.	In an older computer lab, a hub may connect several computers for basic local network communication.
Access Point (AP)	Provides wireless network access to Wi-Fi-enabled devices.	A college access point allows students to connect smartphones and laptops to campus Wi-Fi for YouTube streaming, browsing, and online learning.

- **Explanation of Network Devices**

1. Router: A router directs data between different networks. It commonly connects a home network to the Internet and allows multiple devices to share the same Internet connection.
2. Switch: A switch connects wired devices within a LAN. It forwards Ethernet frames to the appropriate destination port, improving network efficiency.
3. Hub: A hub sends incoming data to all its ports instead of forwarding it only to the intended destination. It is an older networking device and is rarely used in modern networks.
4. Access Point: An access point allows wireless devices, such as smartphones and laptops, to connect to a wired network through Wi-Fi.

❖ **3. Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).**

➢ **Scenario: Ordering Food on Zomato**

- Suppose I am at home and want to order food using the Zomato application on my smartphone. When I open the app, select a restaurant, and place an order, my phone sends data through the home network to the Internet.

➢ **Role of Network Devices**

➢ **Step 1: Smartphone**

I select the food items, enter the delivery address, and tap the *Place Order* button. My smartphone creates a data request containing the order information.

➢ **Step 2: Access Point (Wi-Fi)**

My smartphone connects wirelessly to the home Wi-Fi network. The wireless access point receives the data from the phone and forwards it to the wired network. In a typical home, the access point is built into the Wi-Fi router.

➢ **Step 3: Switch**

If the home network has a separate Ethernet switch, it forwards the data between connected network devices, such as the access point and router. A switch is not always necessary in a simple home network.

- **Step 4: Router**
  The router forwards the data from the home network towards the Internet. It sends the request to the appropriate next network destination, usually through the Internet Service Provider (ISP).

- **Step 5: Internet and Zomato Server**
  The data travels through the ISP's network and the Internet to Zomato's servers. The server processes the order request and sends a response containing order confirmation and other details.

- **Step 6: Response to Smartphone**
  The response returns through the Internet and home router to the Wi-Fi access point. The access point transmits the data wirelessly to my smartphone, where I can see the order confirmation and delivery status.

❖ **4. Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Think about devices like switches, powerline adapters, or WiFi extenders.**

- **Introduction**
  - A gaming PC with an Ethernet port can connect to the Internet through a wired network connection. If the PC is far away from the Wi-Fi router, additional networking devices can help establish a connection without requiring a long Ethernet cable running directly to the router.

- **Suitable Network Devices**

- **1. Powerline Adapter**
- **Working:**
- A powerline adapter uses the existing electrical wiring in a building to carry network data.
  - Connect the first powerline adapter to the Wi-Fi router using an Ethernet cable.
  - Plug the first adapter into a wall electrical socket.
  - Plug the second adapter into a wall socket near the gaming PC.
  - Connect the second adapter to the gaming PC using an Ethernet cable.
- **Advantages:** It provides a wired Ethernet connection without requiring a long cable through the house. Its performance depends on the quality and layout of the electrical wiring.

- **2. Wi-Fi Extender with an Ethernet Port**
- **Working:**
- A Wi-Fi extender receives the wireless signal from the router and extends network coverage to areas with a weak signal.
  - Place the Wi-Fi extender at a location where it receives a good signal from the router.
  - Connect the extender to the router's Wi-Fi network.
  - Connect the gaming PC's Ethernet port to the extender using an Ethernet cable.
- **Advantages:** It allows a PC without built-in Wi-Fi to access the Internet through its Ethernet port. However, connection speed and latency depend on the extender's wireless link and signal strength.